

Set 1: Elementary Group Theory

Problem 1 : Cube power map

Suppose $(xy)^3 = x^3y^3$ for all $x, y \in G$, and G has no elements of order 3. Prove G is abelian.

Tier: ★★ / Skill: power maps / forcing commutativity / Related quals: no close analogue; same flavor as Spring 2007 3, Fall 2017 2

Proof: We will prove this result as a series of claims. The first claim is that $(yx)^2 = x^2y^2$.

$$\begin{aligned} (xy)^3 &= x^3y^3 \\ \Rightarrow xyxyxy &= x^3y^3 \\ \Rightarrow (yx)^2 &= x^2y^2. \end{aligned}$$

Consider $p : G \rightarrow G$ where $x \rightarrow x^3$, the assumption about the group tells us that p is a homomorphism furthermore the fact that no element has order 3, p is injective. Suppose

$$x^3y^3 = y^3x^3 \Rightarrow p(xy) = p(yx) \Rightarrow xy = yx.$$

Thus if the cubes of two elements commute then the two elements themselves commute. Now we will prove that

$$x^3y^2 = y^2x^3$$

for any pair x and y .

$$\begin{aligned} x^3y^2 &= xx^2y^2 \\ &= xyxyx \\ &= (xy)^2x \\ &= y^2x^2x = y^2x^3. \end{aligned}$$

Now we will prove that everything commutes with squares.

$$x^3(y^2)^3 = (y^2)^3x^3 \quad (x^3y^2 = y^2x^3 \text{ repeatedly}).$$

Thus $xy^2 = y^2x$.

$$\begin{aligned} yxyx &= (yx)^2 \\ &= x^2y^2 \\ &= y^2x^2 \\ xy &= yx. \end{aligned}$$

□

Problem 2 : Central quotient and the 5/8 theorem

Prove that if $G/Z(G)$ is cyclic, then G is abelian. Then prove the “5/8 theorem”: if G is finite and nonabelian, the probability that two random elements commute is at most $\frac{5}{8}$.

Tier: ★ (first claim), ★★ (5/8) | Skill: centers, class equation / counting commuting pairs | Related quals: Fall 2001 G1, Fall 2004 Problem 1, Spring 2001 A2, Fall 2011 Problem 2

Proof: Suppose $G/Z(G)$ is cyclic. Thus exists $g \in G$ such that $G/Z(G) = \langle gZ(G) \rangle$. Thus for all $a, b \in G$, $a = g^k z_1$ and $b = g^m z_2$. Hence

$$\begin{aligned} ab &= g^k z_1 g^m z_2 \\ &= g^m z_2 g^k z_1 \\ &= ba. \end{aligned}$$

Thus G is abelian. □

Problem 3 : Unions of subgroups

Prove that no group is a union of two proper subgroups. Find a group that is a union of three. For a challenge, characterize all such groups.

Tier: ★★ (challenge ★★★) | Skill: covering arguments, counting via quotients | Related quals: Spring 2016 9, Spring 2004 Problem 1

Proof: Assume for the sake of contradiction, $G = H \cup K$ where H, K are proper. Suppose $h \in H - K$ and $k \in K - H$. $hk \in G$. We have various cases:

1. $hk \in H$: Then $h^{-1}hk = k \in H$ which is a contradiction.
2. $hk \in K$: Then $h = hkk^{-1} \in K$, which is a contradiction.

Thus $hk \notin H \cup K = G$, thus a contradiction occurs. □

Problem 4 : Smallest prime index

Let p be the smallest prime dividing $|G|$. Prove that a subgroup of index p is normal. (Hint: let G act on the cosets.)

Tier: ★★ | Skill: action on cosets, $G \rightarrow S_p$ | Related quals: Spring 2004 Problem 3, Fall 2014 6, Winter 2005 A1, Fall 2006 G2

Proof:

□

Problem 5 : Finite subgroups of a field's units

Prove that every finite subgroup of the multiplicative group of a field is cyclic. (Count elements of each order and compare with $\sum_{d|n} \varphi(d) = n$.)

Tier: ★★ | Skill: counting elements by order; $\sum_{d|n} \varphi(d) = n$ | Related quals: Fall 2000 A3, Fall 2008 G2

Problem 6 : Generating S_n

Prove S_n is generated by $(1\ a)$ and $(1\ 2\ \cdots\ n)$ if and only if $\gcd(a-1, n) = 1$.

Tier: ★★ / Skill: generators of S_n , transitivity plus transposition / Related quals: Spring 2008 G1, Winter 2003 A3

Problem 7 : Generators of $\mathrm{SL}_2(\mathbb{Z})$

Prove $\mathrm{SL}_2(\mathbb{Z})$ is generated by $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. (Hint: think of the Euclidean algorithm.)

Tier: ★★ / Skill: Euclidean algorithm as row reduction / Related quals: Spring 2006 G2, Fall 2006 G3, Spring 2020 6 (finite-field SL_2 , related setting)

Problem 8 : Subgroups of $\mathbb{Q}\mathbb{Q}\mathbb{Q}$

Prove that every finitely generated subgroup of $\mathbb{Q}$ is cyclic, that $\mathbb{Q}$ has no maximal subgroups, and that $\mathbb{Q}/\mathbb{Z}$ has exactly one subgroup of each finite order.

Tier: ★★ / Skill: divisible groups, f.g. torsion-free rank 1 / Related quals: Winter 2005 A2, Fall 2008 G3, Fall 2005 Sec.1, Spring 2009 G2

Problem 9 : $\mathrm{Aut}(\mathbb{Z}/n)$ and groups of order 6

Compute $\mathrm{Aut}(\mathbb{Z}/n)$. Show $\mathrm{Aut}(\mathbb{Z}/2 \times \mathbb{Z}/2) \cong S_3$, and use this to classify the groups of order 6 by hand.

Tier: ★ / Skill: automorphism groups, semidirect products / Related quals: Spring 2012 10, Fall 2006 G1, Winter 2002 A2, Fall 2013 1, Spring 2015 8, Spring 2008 G2

Set 2: Category Theory and Groups

Problem 10 : Epimorphisms in $\mathbf{Grp}$

Prove that epimorphisms in $\mathbf{Grp}$ are surjective. By contrast, show that the inclusion $\mathbb{N} \hookrightarrow \mathbb{Z}$ is an epimorphism in $\mathbf{Mon}$.

Tier: ★★★ / Skill: universal properties, constructing test maps out of G / Related quals: Spring 2015 1, Spring 2015 2

Problem 11 : Free groups on finite sets

Prove that if $F(A) \cong F(B)$ as groups, then $|A| = |B|$ for finite sets A and B . (Hint: count homomorphisms to $\mathbb{Z}/2$.)

Tier: ★★ / Skill: counting homomorphisms out of a free object / Related quals: Fall 2016 1, Spring 2017 3, Winter 2005 A3

Problem 12 : Coproducts

Show that $\mathbb{Z}/2 * \mathbb{Z}/2$ (the coproduct in $\mathbf{Grp}$) is the infinite dihedral group. Then explain why the coproduct in $\mathbf{Ab}$ is so different.

Tier: ★ / Skill: free products, presentations / Related quals: Spring 2015 1, Fall 2016 1, Spring 2020 1

Problem 13 : Fibered products

Describe fibered products in **Set** and **Grp**, and show that the kernel of a homomorphism is one of them.

Tier: ★ / Skill: limits as universal objects / Related quals: Spring 2015 2 (categorical flavor)

Problem 14 : Representable functors on groups

Show that $\text{Hom}_{\mathbf{Grp}}(\mathbb{Z}, -)$ is naturally isomorphic to the forgetful functor. What represents the functor sending G to its set of elements of order dividing n ?

Tier: ★★ / Skill: Yoneda, representability / Related quals: Spring 2011 Problem 10

Problem 15 : Mono, epi, and isomorphism

Find a category where a morphism is both mono and epi but not an isomorphism. Then prove that a morphism that is mono and split epi is an isomorphism.

Tier: ★ / Skill: basic categorical morphism types / Related quals: Spring 2011 Problem 10